International and Monetary Economics

Course Code: 30054

Professor: Dmitriy Sergeyev

Lecture 7

Roadmap

1. Money and prices in the short run
   - Exercises
2. Money and prices in the long run
   - Monetary neutrality
   - Evidence
3. Money and FX in the long run
   - Monetary neutrality
   - Exercise

The Price of War

What is the macroeconomic impact of war?

- Aggregate demand
  - Fiscal: military buildups (G), fiscal (T , T^e) and monetary (M) financing
  - Consumption (C), investment (I), and trade (Ex, Im) collapse
- Aggregate supply
  - Labor: death, immigration, decline in human capital
  - Capital: destruction
  - Sectoral disruptions: supply chain disruptions, commodities shortages
- And more

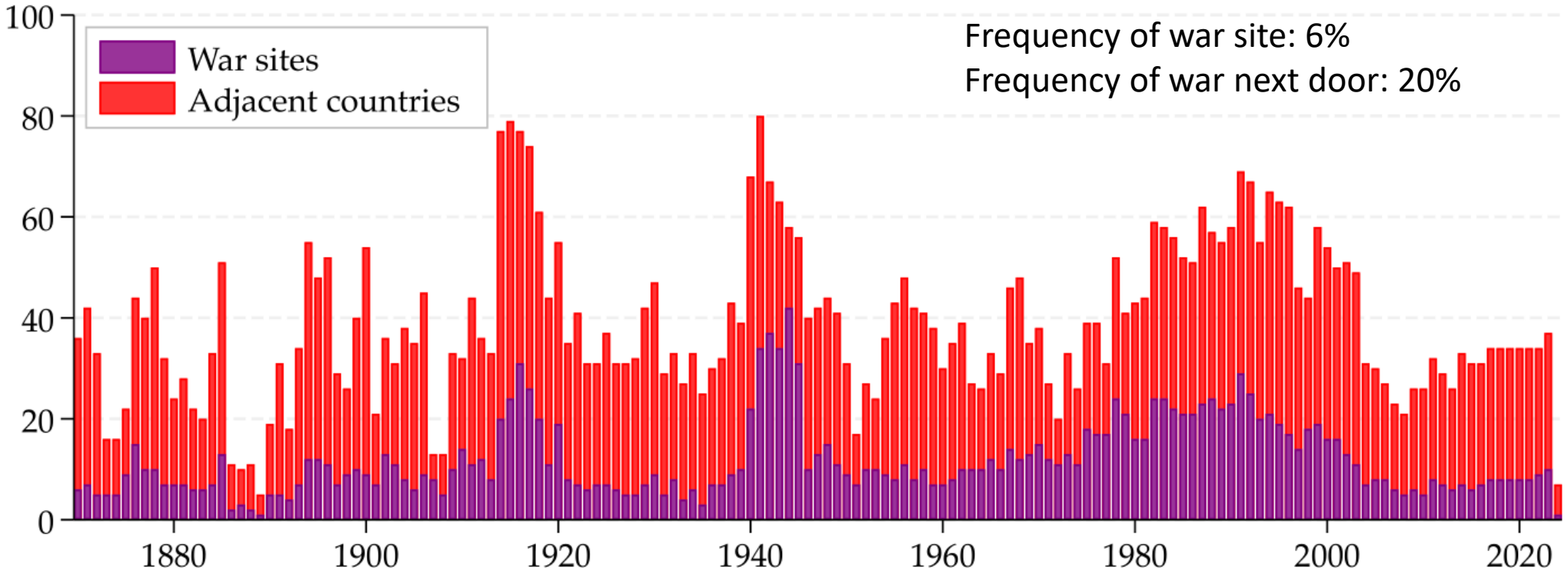
Historical Perspective

- Paper: Federle, Meier, Müller, Mutschler, Schularick (2026, AER), "The Price of War"
- Data
  - 150 years
  - 60 countries
  - Data on war sites, parties, trade links, etc.
- Main econometric specification

$$y_{i,t+h} - y_{i,t-1} = \zeta_h \mathbf{X}_{i,t} + \sum_{c \in \{S,B,T\}} \phi_{c,h}(\gamma'_{i,c,t} \cdot \mathbf{Sites}_t) + \sum_{c \in \{B,T\}} \psi_{c,h}(\epsilon'_{i,c,t-1} \cdot \mathbf{Sites}_t) + u_{i,t+h}.$$

  - see the paper on detailed discussion of what all these terms mean
  - **Why is it ok to estimate such a regression?** Countries rarely go to war because of the business cycle considerations (there are exceptions, however)

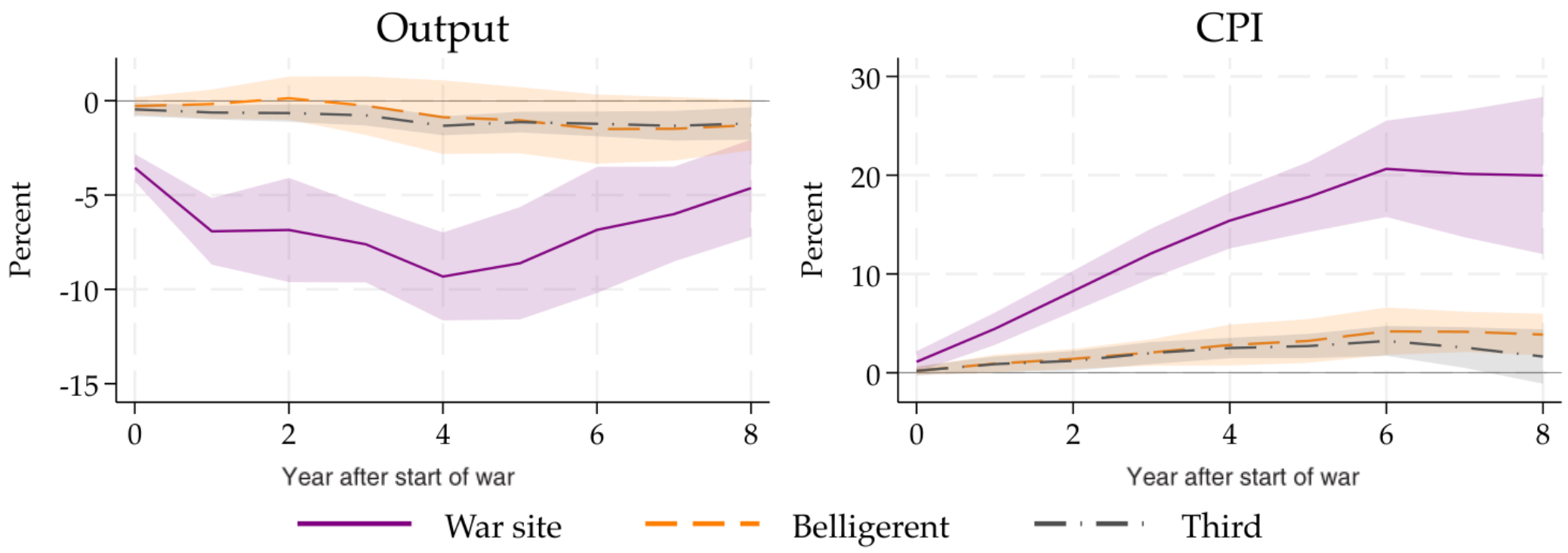
Countries exposed to war 1870–2023

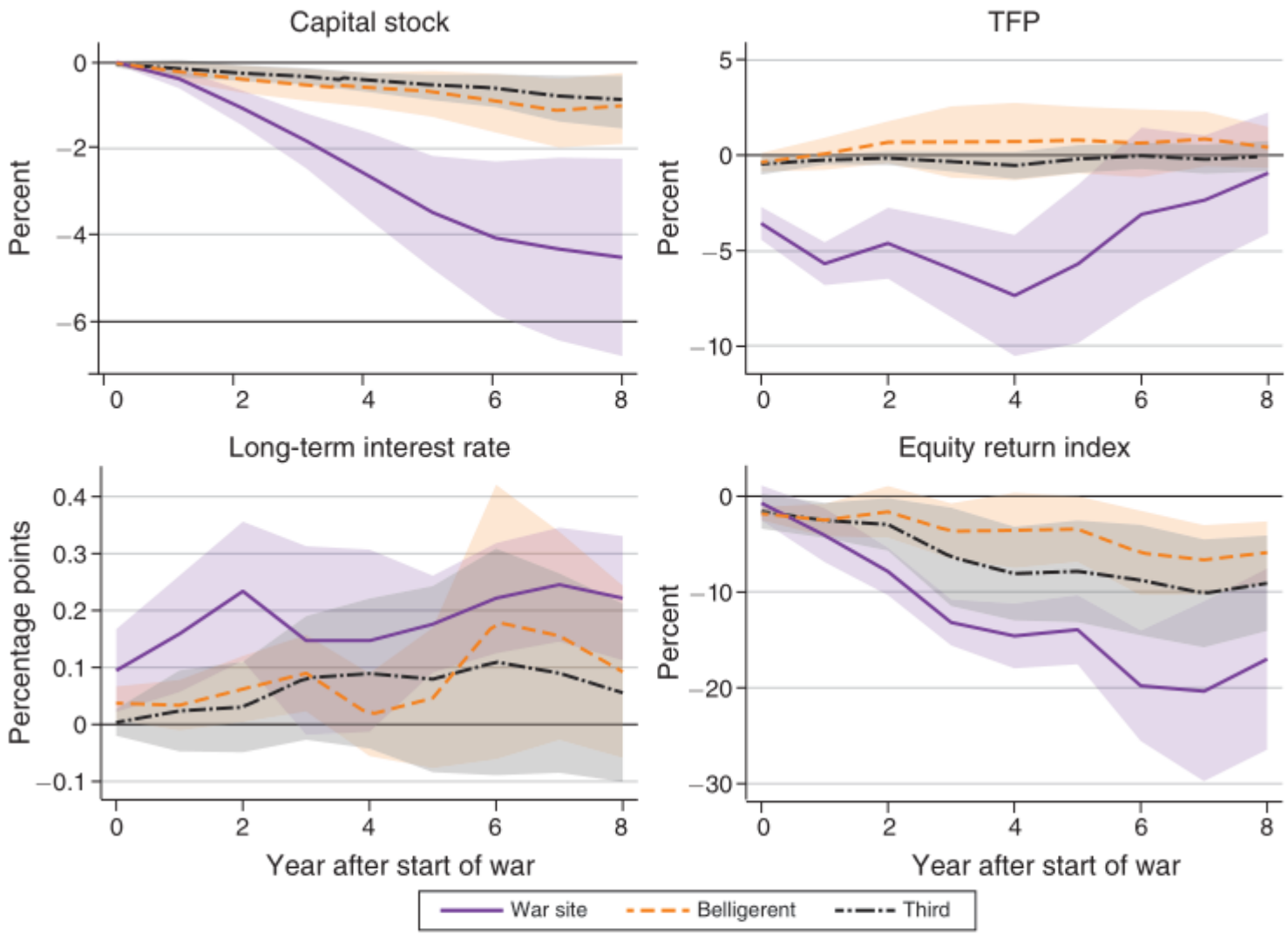


Source: «The Price of War», AER, 2026

Strong adverse effect on the war site, small spillovers on average

Severity 2 percent (casualties/population); size of war site 3 percent (population/world population)



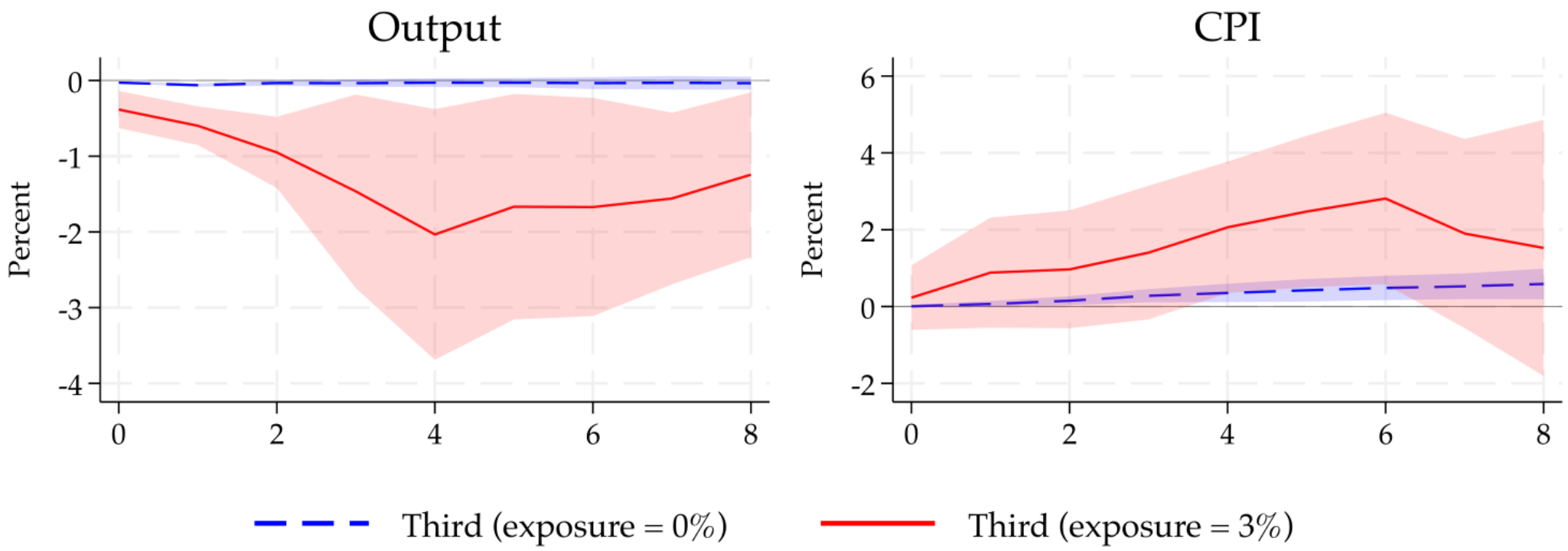


1. Capital and equity returns decline persistently despite the fact that the median war duration is 2 years
   - Why?

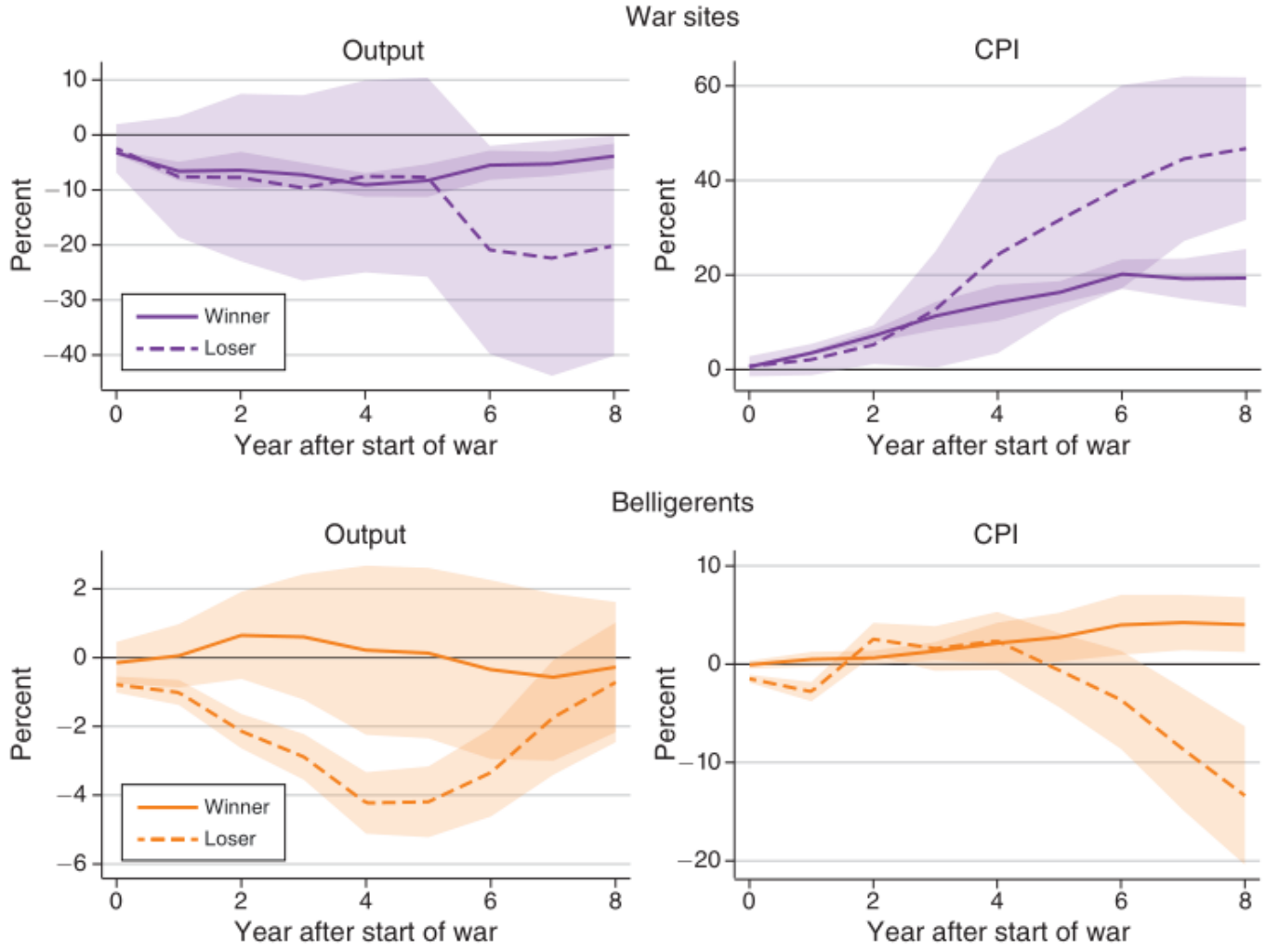
2. TFP rebounds after a few years

Strong adverse spillovers on third countries if close to war site

Severity 2 percent (casualties/population); scenarios of different trade exposure (imports from sites/gdp)



Losers suffer more than winners



Money and FX Markets Equilibrium
(in the short run)

Money and FX Markets Equilibrium

- FX market

$$R = R^* + \frac{E^e - E}{E}$$

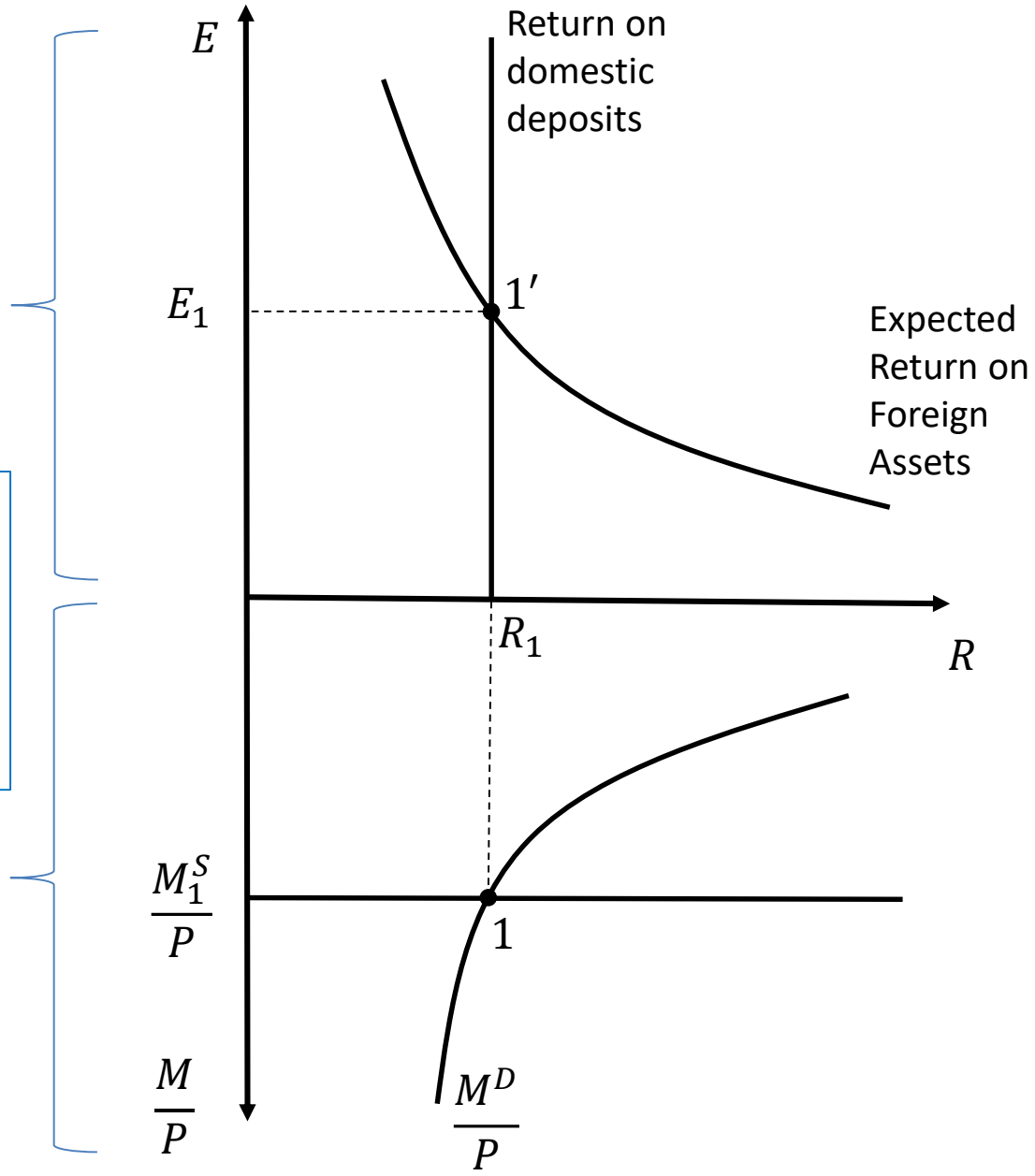
- Money market (at home)

$$\frac{M^s}{P} = L(R, Y)$$

To proceed we need to know how aggregate output (Y), the price level (P), and the future exchange rate (E^e) are determined!

- Assumptions
  1. Y, Y^* are fixed **for simplicity**
  2. E^e is fixed **for simplicity** (because perhaps policy experiments are ***temporary*** and do not affect future exchange rate)
  3. P, P^* are fixed **in the short run**

Simultaneous Equilibrium in the home Money and the FX Markets



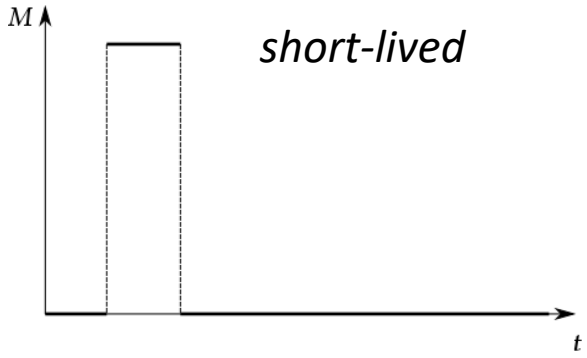
Exercise 1

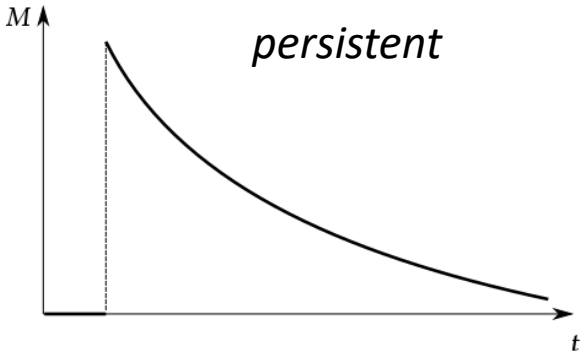
An Increase in domestic money supply

- How does the current exchange rate change when a domestic central bank **temporarily** increases its money supply?

Language

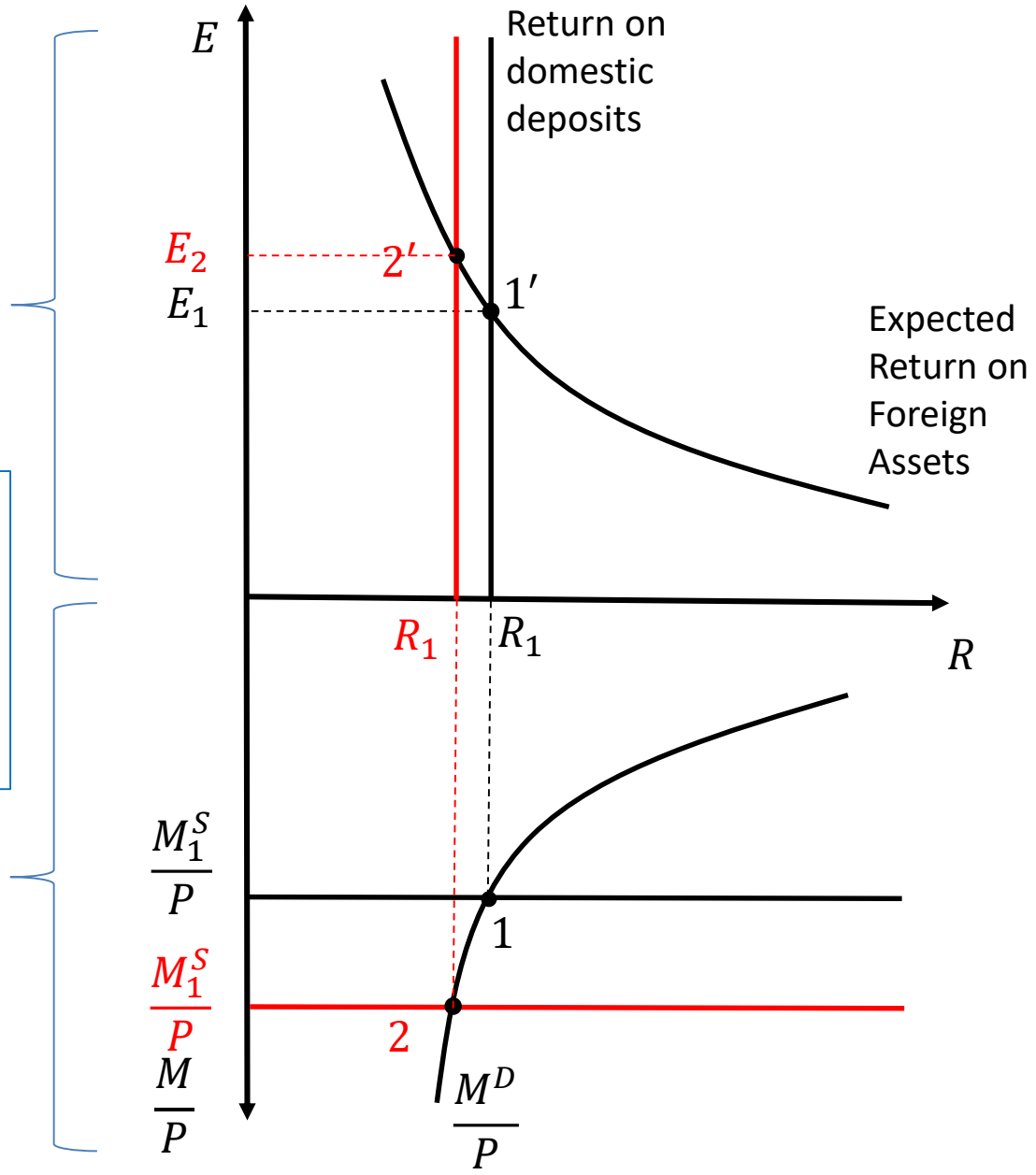
- A temporary change





- A permanent change

Simultaneous Equilibrium in the home Money and the FX Markets



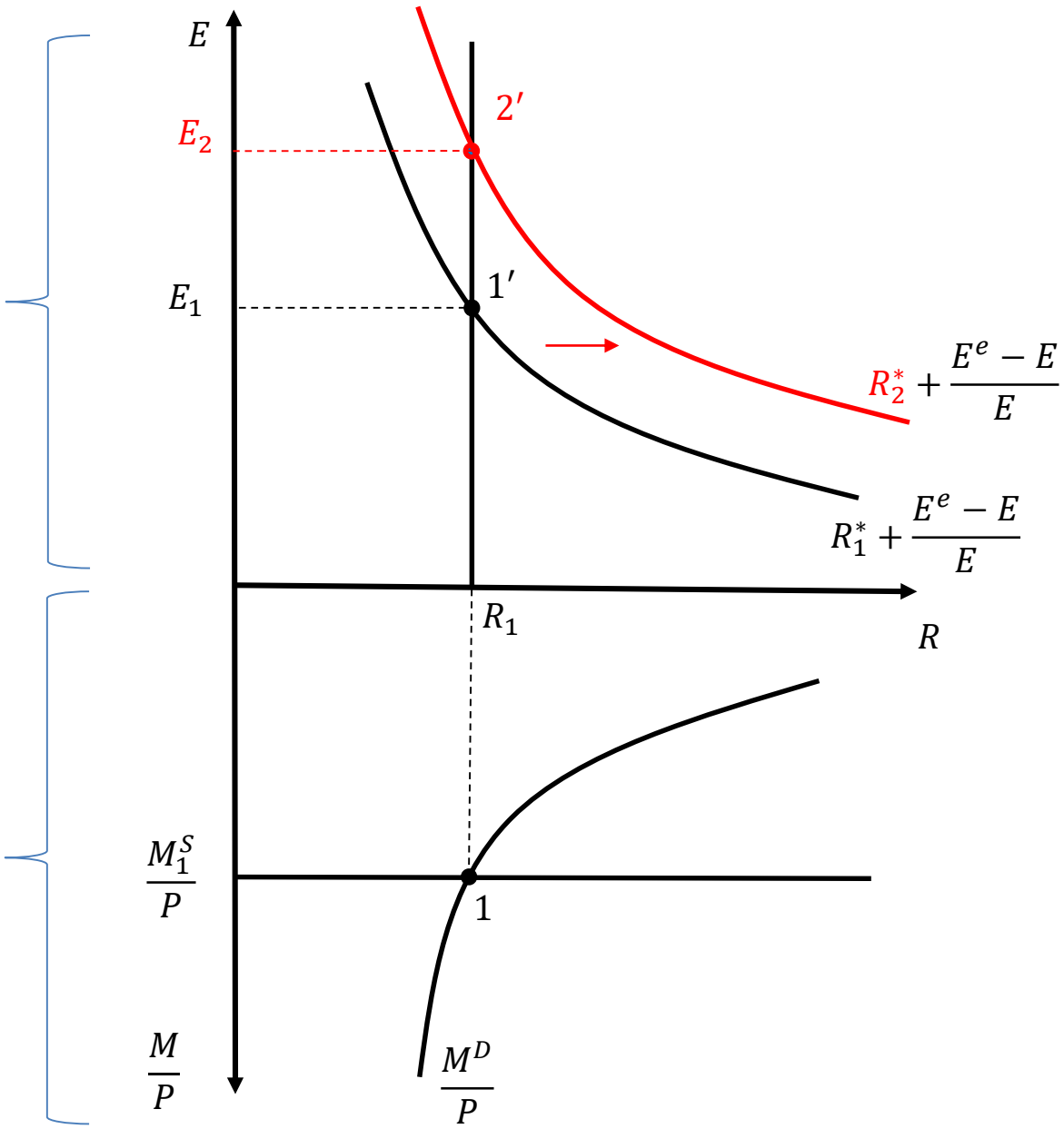
Exercise 2

A Decrease in foreign money supply

- How does the current exchange rate change when foreign money supply shrinks temporarily?

Change in foreign monetary policy

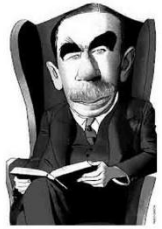
$$M^{*S} \downarrow \Rightarrow R^* \uparrow$$



Money and Prices

(in the long run)

Short Run vs. Long Run



- In the short run
  - Assumptions: (i) **prices are sticky**, (ii) output is fixed (for simplicity)
  - The money market equilibrium determines the nominal interest rate

- In the long run
  - Assumptions: **prices are flexible**; output vary but it is determined by real factors (as opposed to monetary policy changes)
  - The money market equilibrium determines the equilibrium price level

Money and Prices in the Long Run

- Money market equilibrium

$$\frac{M^s}{P} = L(\bar{r} + \bar{\pi}, \bar{Y})$$

  - $\bar{Y}$ is the level of output determined by real factors (labor force, capital stock, technology, etc.)
  - $\bar{r}$ is the real interest rate determined by real factors (trend output growth, population growth, etc.)
  - $\bar{\pi}$ is the long-run inflation (inflation target by the central bank)
  - **Bars on top of the variables** mean that the variables are **exogenous**. In this case, it means that they do not depend on the level of money supply

Money and Prices in the Long Run

- Long-run monetary neutrality (proposition)
  - **Real output**: a change in the level of the supply of money does not affect the long-run value of real output
  - **Price level**: a *permanent* increase in the money supply causes a proportional increase in the price level's long-run value

- Changes in money demand also affect the long-run price level
  - $P = M^s/L(\bar{R}, \bar{Y})$ which implies $\Delta P/P \approx \Delta M^s/M^s - \Delta L/L$

Empirical Evidence

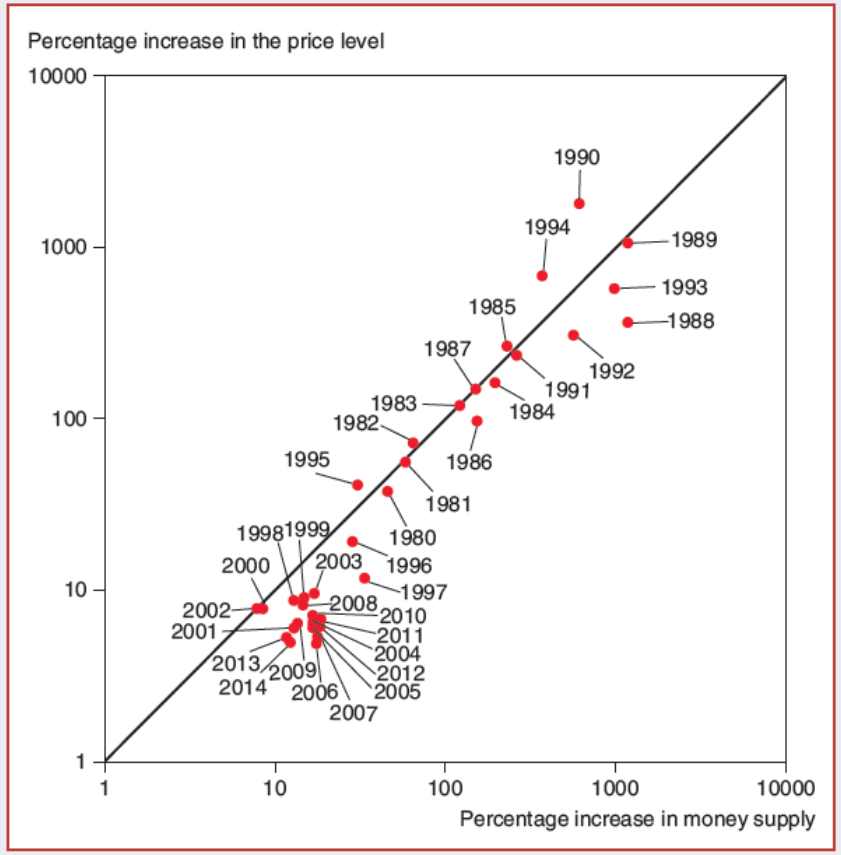


FIGURE 15-10

Average Money Growth and Inflation in Western Hemisphere Developing Countries, by Year, 1980–2014

Even year by year, there is a strong positive relation between average Latin American money supply growth and inflation. (Both axes have logarithmic scales.)

Source: World Bank development indicators database and own calculations. Regional aggregates are weighted by shares of dollar GDP in total regional dollar GDP.

1. Money supply and price level
   Over a long period (1980-2014), changes in the price level are highly correlated with changes in the money supply.

2. Money demand and price level
   The deviations from the 45-degree line suggest that changes in money demand also affect the price level.

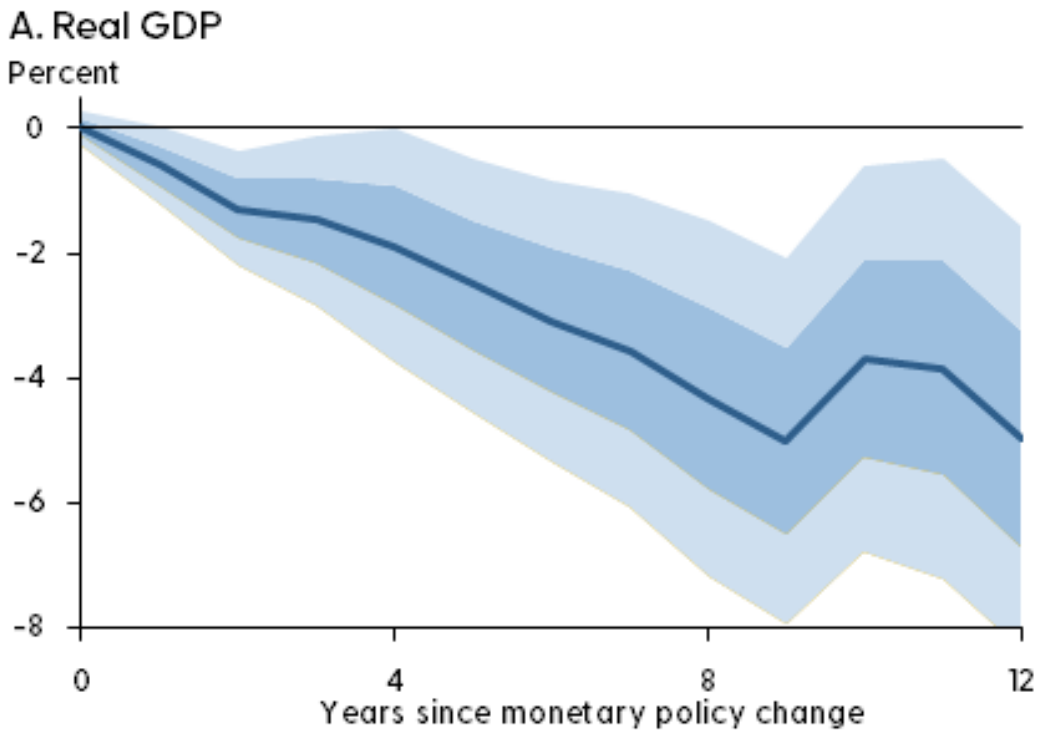
Is Money Really Neutral in the Long Run?

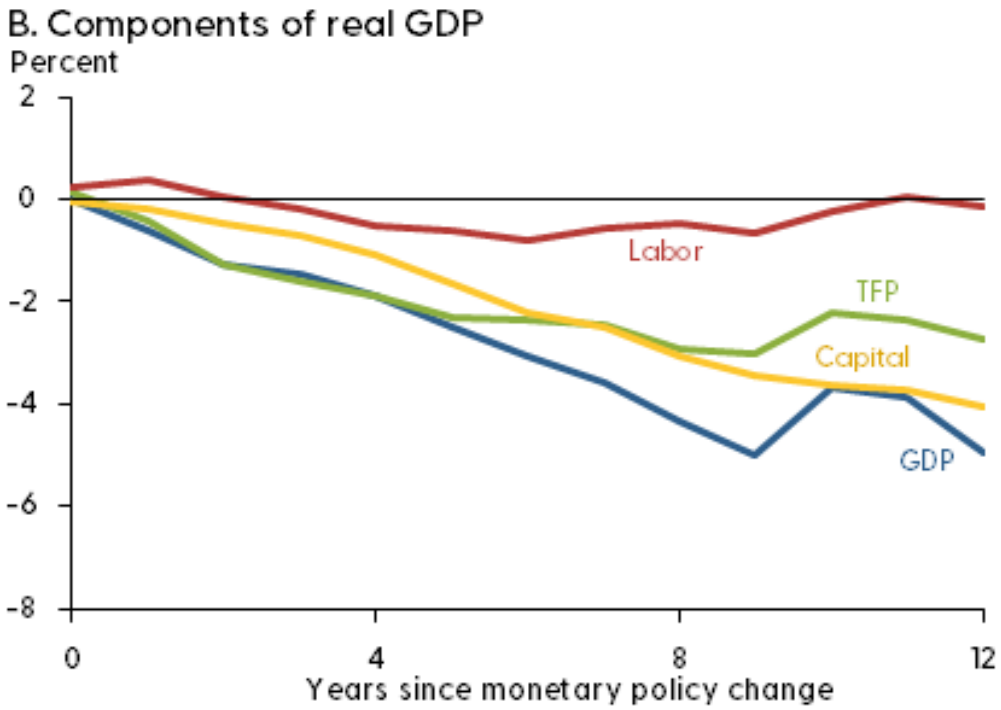
Effects of Monetary Policy in the Long Run

- Why can one expect money ***not*** to be neutral in the long run?
  - R&D reaction to interest rate changes
  - Human capital depreciation during recessions

- Jordà, Singh, Taylor (2023) test the long-run neutrality of MP
  - Historical data: 17 advanced economies over the period 1900 to 2015, excluding the two World Wars
  - Need a good identification of monetary policy shocks
    - Identification strategy: countries that peg their currencies to the US dollar must react to the US monetary policy variations unrelated to economic activity in those countries

Jordà, Singh, Taylor (2023)





Note: Responses to a 1% unexpected change in domestic short-term interest rate. Full sample: 1900–2015, World Wars excluded.
In panel A, darker shading indicates one-standard-error and lighter shading indicates two-standard-error confidence bands around average.
See Jordà, Singh, and Taylor (2023) for estimation details

Evidence on the Innovation Channel

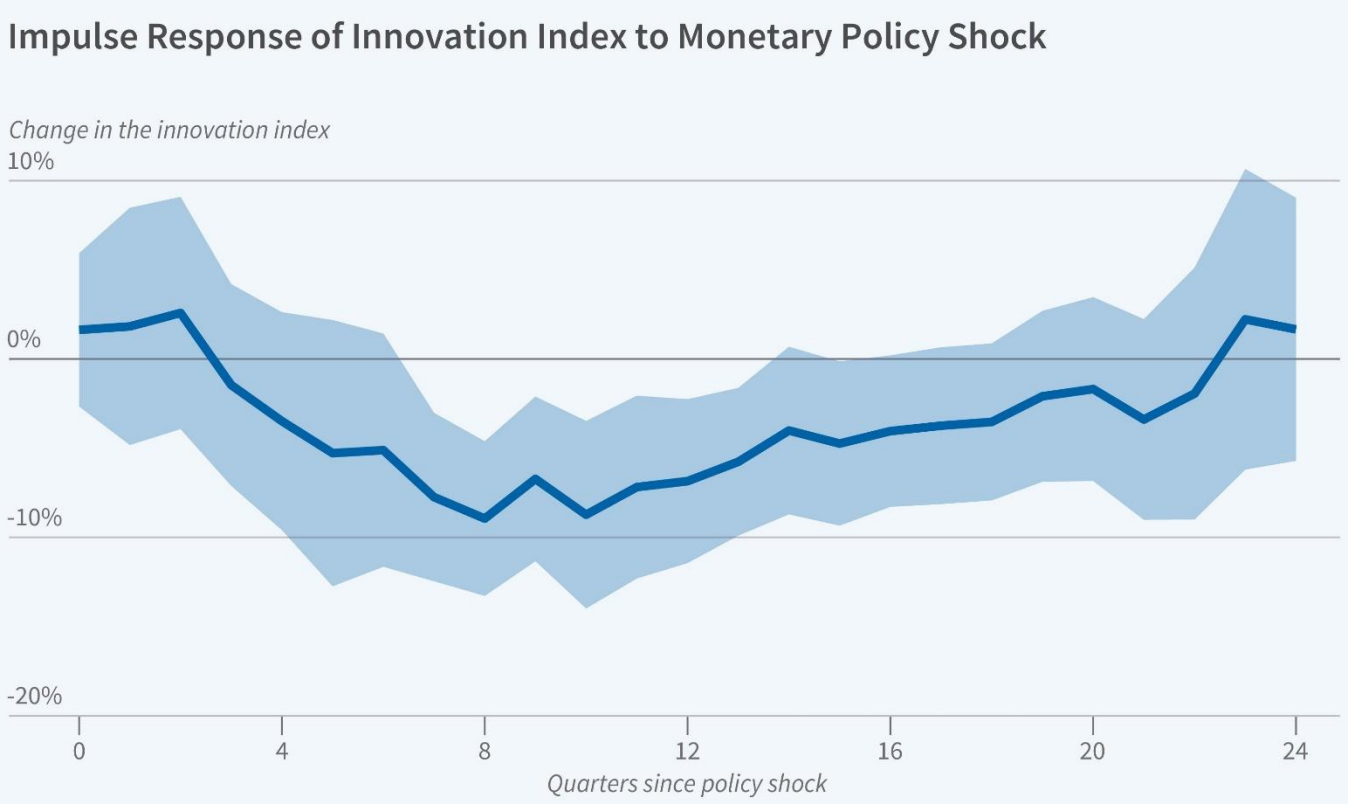


Figure shows response to a 100 basis point interest rate increase from a monetary policy shock. Shaded area represents 90% confidence intervals.
Source: Researchers' calculations using data from NBER Working Paper w17769.

Source: Ma, Zimmermann (2023)

Money and FX

(in the long run)

Money and FX in the Long Run

- Long-run monetary neutrality (proposition) **again**
  - a permanent increase in a country's money supply causes a *proportional* long-run depreciation of its currency against foreign currencies

$$E = \frac{P}{P^*}$$

- Why?
  - Purchasing power parity theory (next time)

Short Run vs. Long Run

- In the short run, when prices (and output) are fixed
  - changes in money supply affect interest rates and, as a result, the exchange rate

$$M \uparrow \Rightarrow \boldsymbol{R} \downarrow \Rightarrow E \uparrow$$

- In the long run, when prices are flexible
  - changes in money supply affect prices and, as a result, the nominal exchange rate

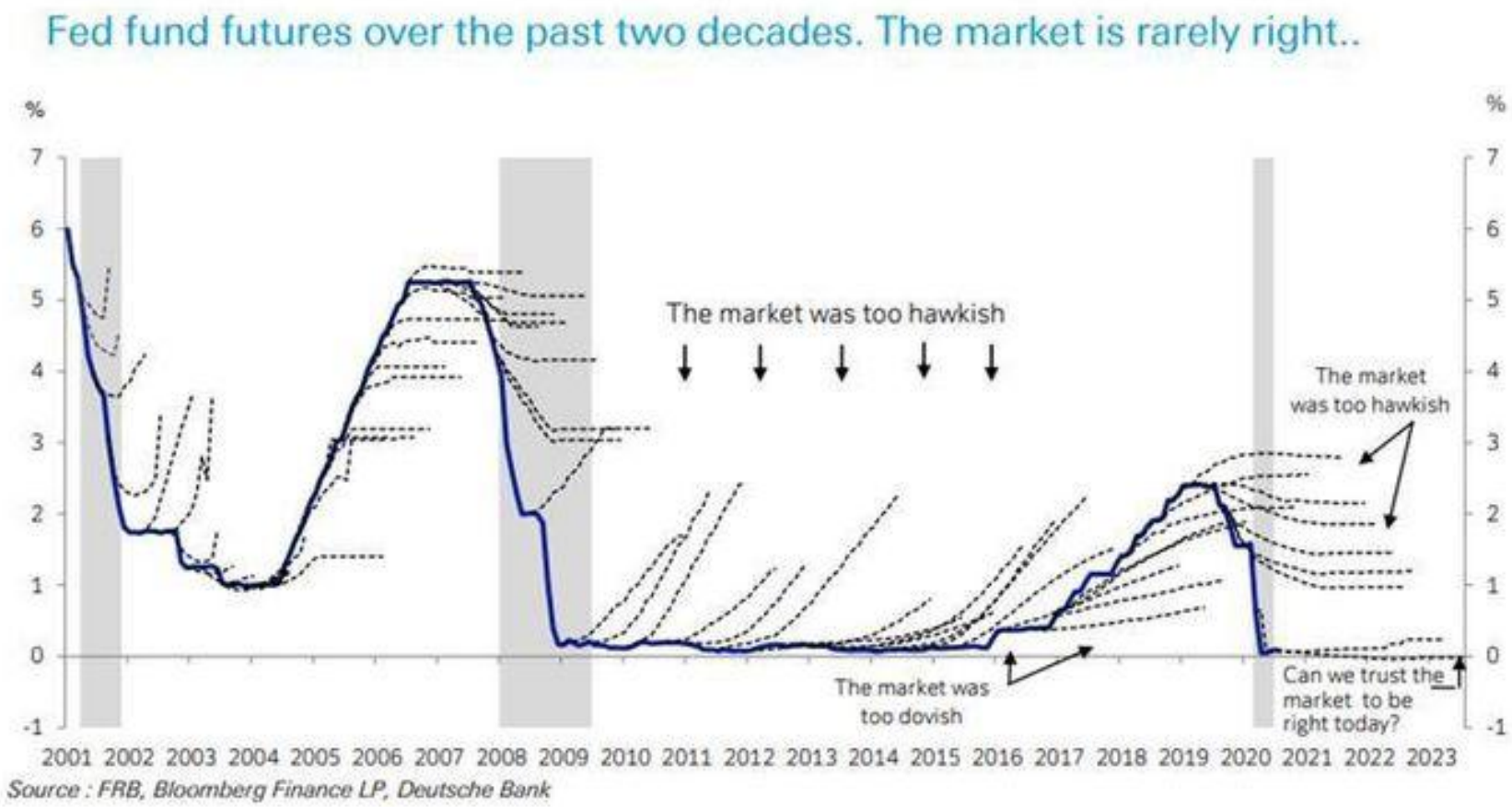
$$M \uparrow \Rightarrow \boldsymbol{P} \uparrow \Rightarrow E \uparrow$$

Exercise

A **permanent** increase in domestic money supply

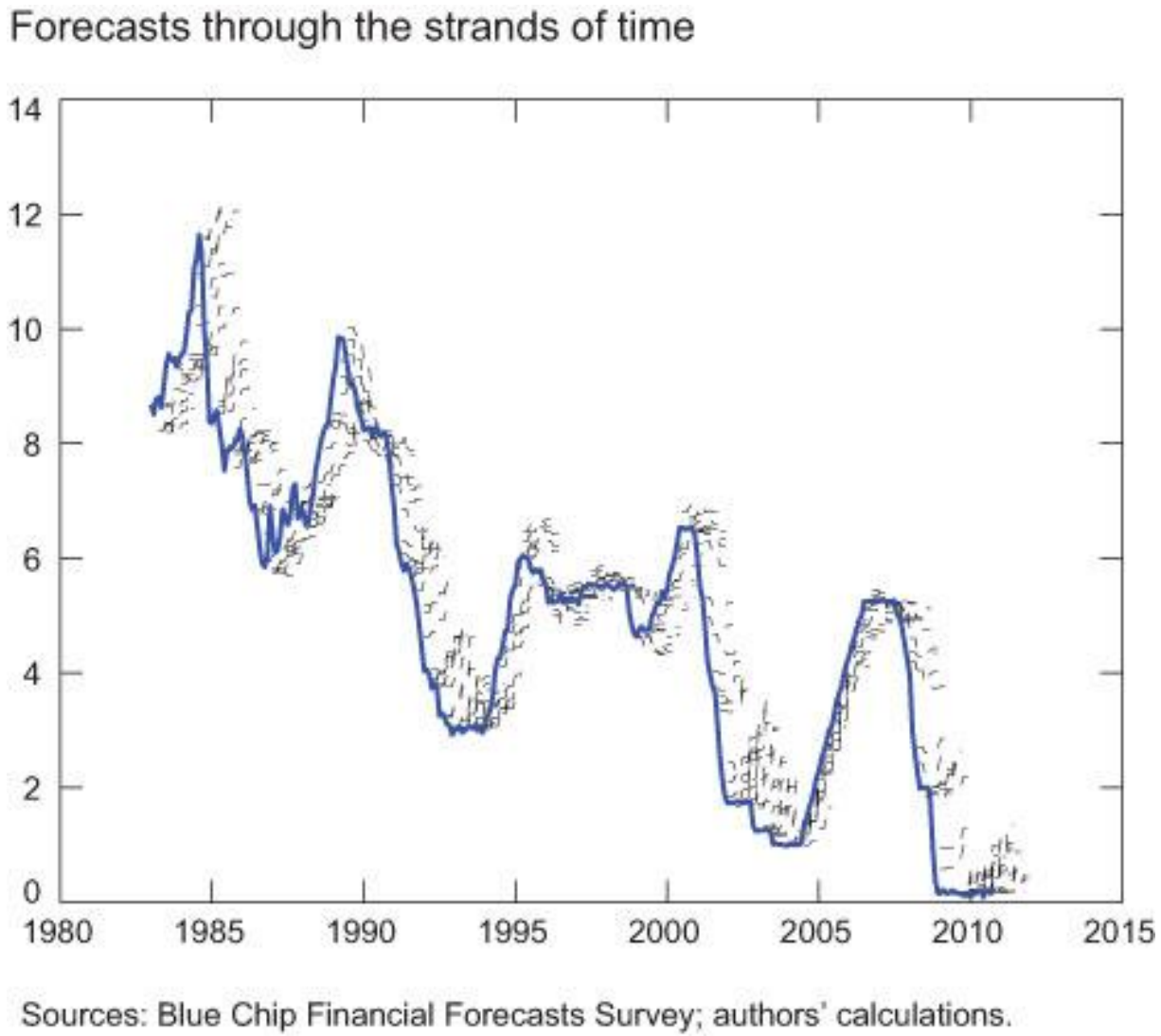
- How does the current exchange rate change when a domestic central bank increases its money supply **permanently**?

Markets Expectations about Monetary Policy



Source: [MarketWatch]

Markets Expectations about Monetary Policy



Source: [Liberty Street Economics]

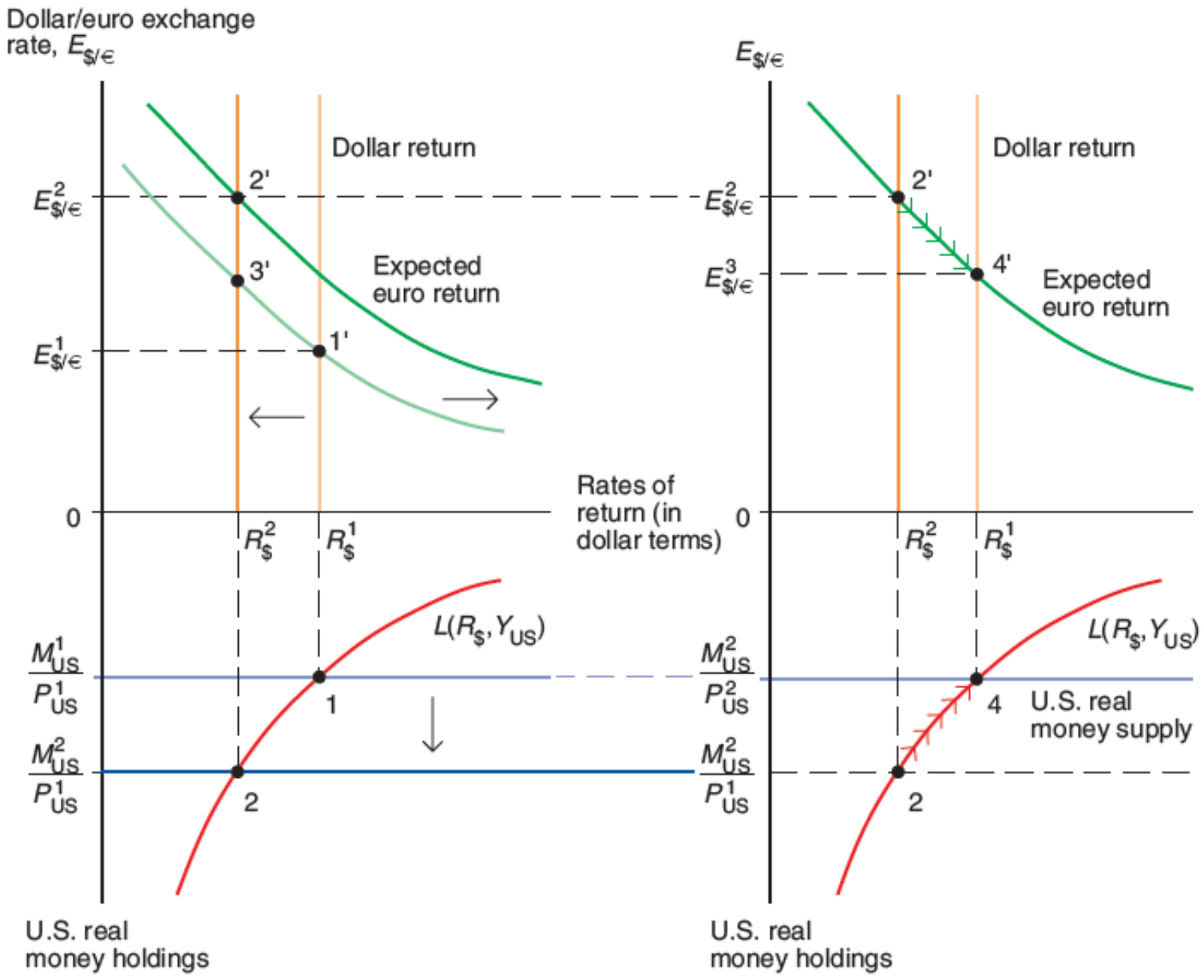
Exercise

A **permanent** increase in domestic money supply

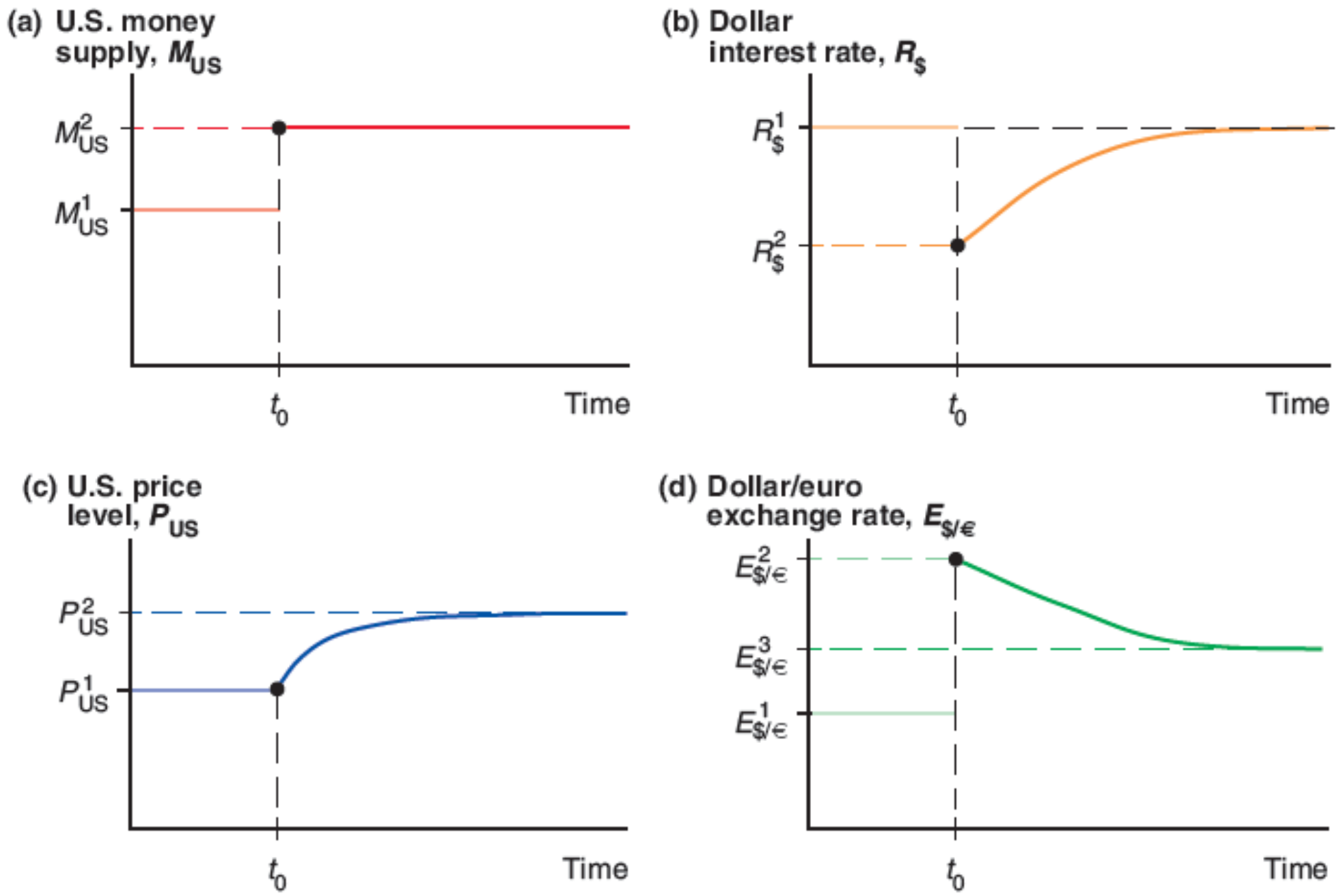
- This exercise combines our short-run and long-run analysis of monetary changes
  - A change in money supply **not only** affects the nominal interest rate and exchange rate like in our previous short-run analysis, **but also** the expected exchange rate

- It is sometimes called

  "the Dornbusch overshooting model"



(a) Short-run effects **(b) Adjustment to long-run equilibrium**


(a) U.S. money supply, M_{US}
M^2_{US}
M^1_{US}
t_0
Time
(b) Dollar interest rate, $R_\$$
$R^1_\$$
$R^2_\$$
t_0
Time
(c) U.S. price level, P_{US}
P^2_{US}
P^1_{US}
t_0
Time
(d) Dollar/euro exchange rate, $E_{\$/€}$
$E^2_{\$/€}$
$E^3_{\$/€}$
$E^1_{\$/€}$
t_0
Time

Exercise Takeaways

A **permanent** increase in domestic money supply

1. **We say that** the exchange rate **overshoots** when its immediate response to a disturbance is greater than its long-run response

2. The model helps explain why exchange rates move so sharply in the short run

3. **Overshooting** is a direct result of a sluggish short-run price level adjustment and the interest parity

Next Time

- KOM, Chapter 16